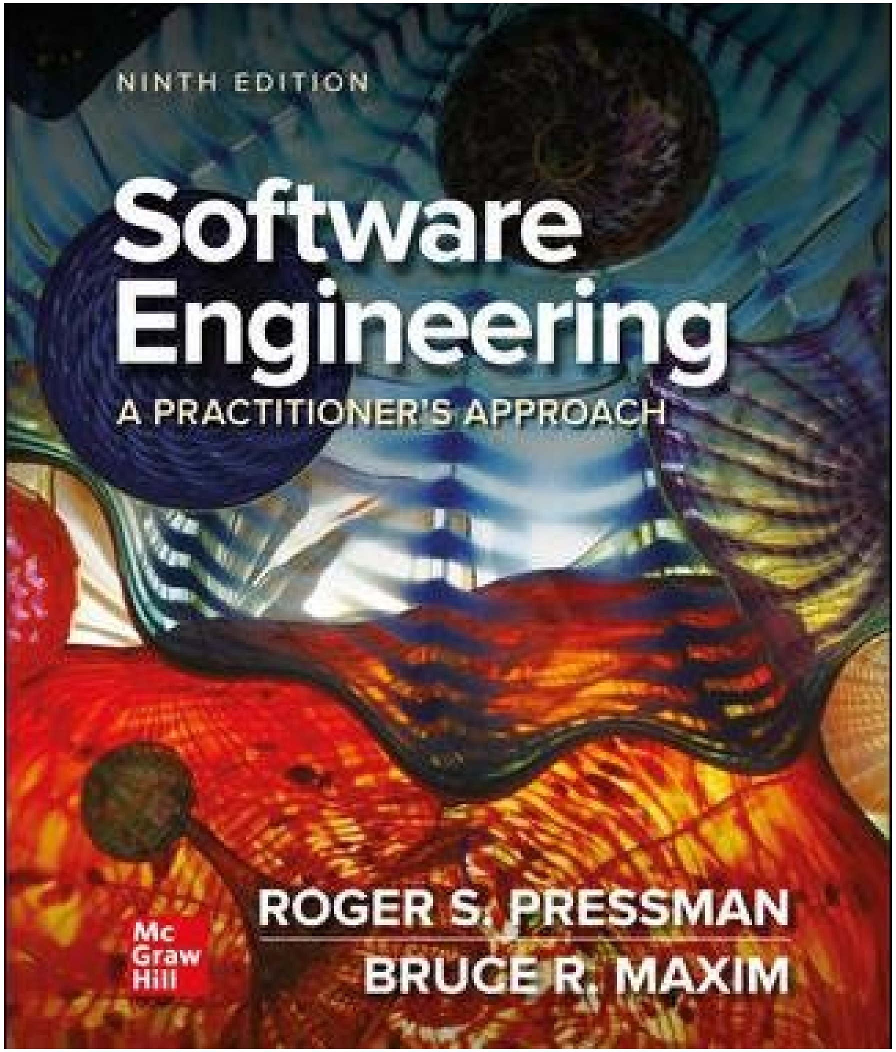
Because learning changes everything.®

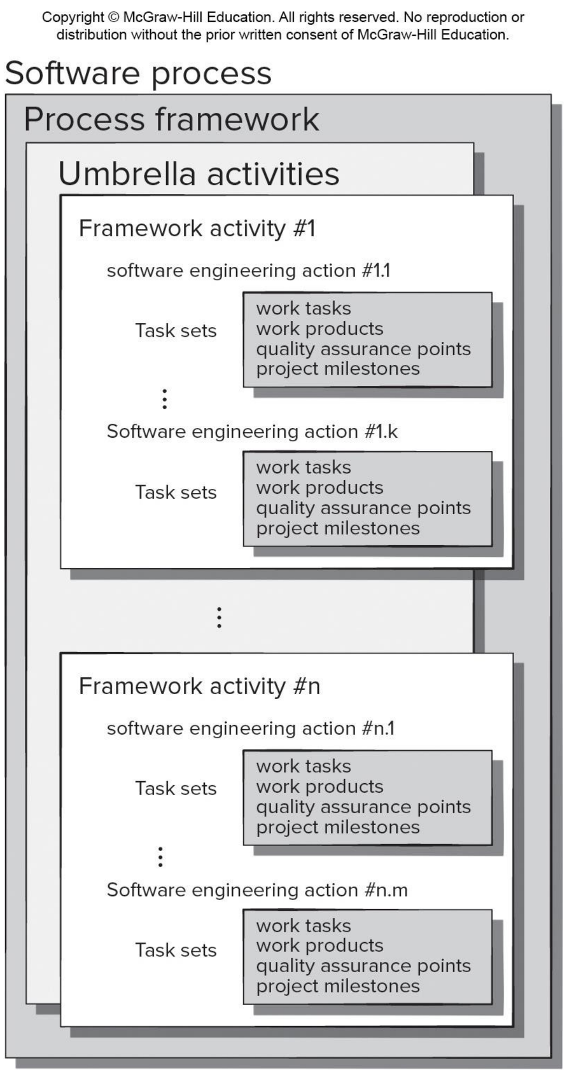
Chapter 2

Process Models

Part 1 - The Software Process



Generic Process Model



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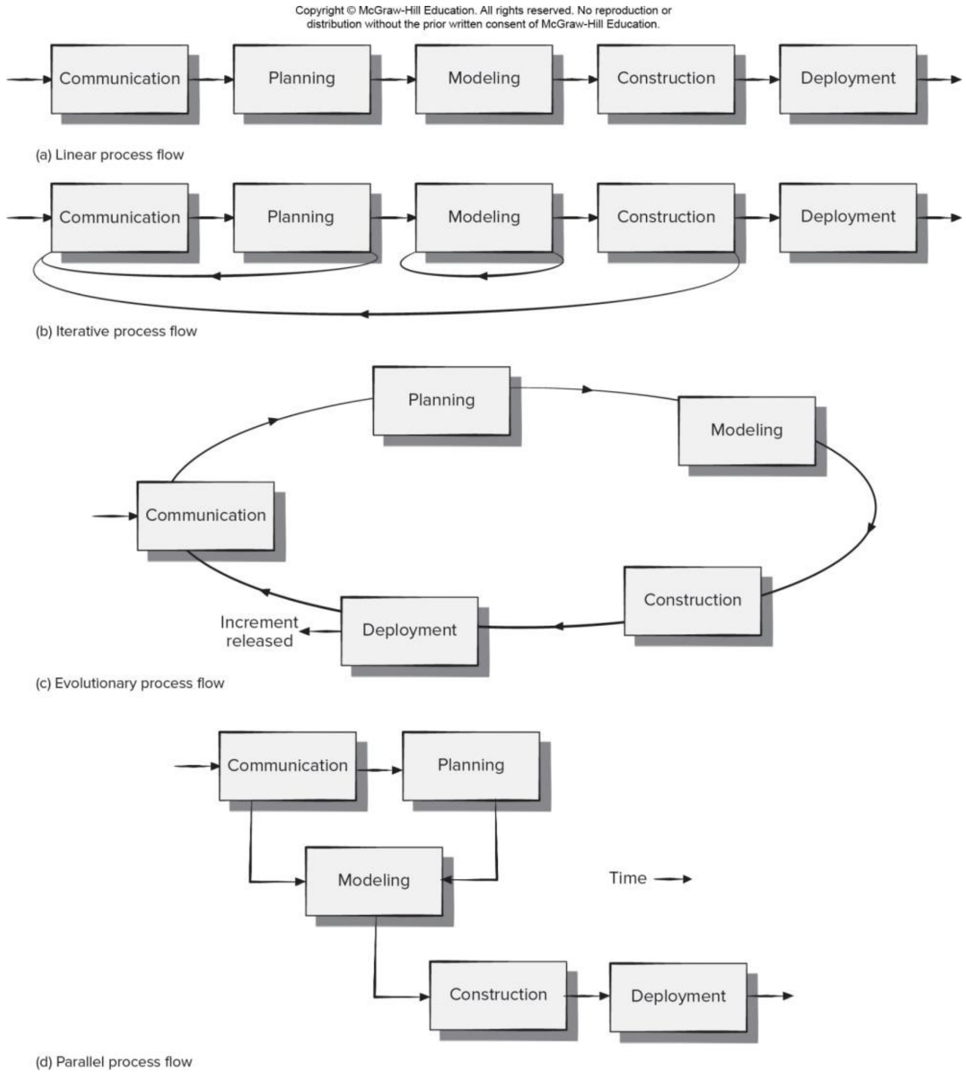
Process Flow



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Identifying a Task Set

A task set defines the actual work to be done to accomplish the objectives of a software engineering action.

A task set is defined by creating several lists:

- A list of the tasks to be accomplished.
- A list of the work products to be produced.
- A list of the quality assurance filters to be applied.

Process Assessment and Improvement

- The existence of a software process is no guarantee that software will be delivered on time, or meet the customer’s needs, or that it will exhibit long-term quality characteristics.
- Any software process can be assessed to ensure that it meets a set of basic process criteria that have been shown to be essential for successful software engineering.
- Software processes and activities should be assessed using numeric measures or software analytics (metrics).

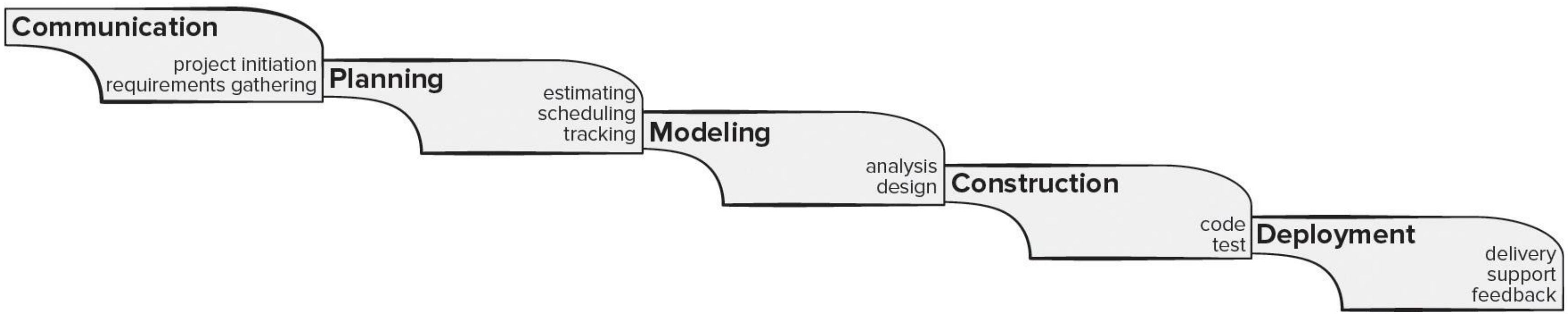
Prescriptive Process Models [1]

Prescriptive process models advocate an orderly approach to software engineering.

That leads to a two questions:

- If prescriptive process models strive for structure and order, are they appropriate for a software world that thrives on change?
- If we reject traditional process models and replace them with something less structured, do we make it impossible to achieve coordination and coherence in software work?

Waterfall Process Model



Pros

- It is easy to understand and plan.
- It works for well-understood small projects.
- Analysis and testing are straightforward.

Cons

- It does not accommodate change well.
- Testing occurs late in the process.
- Customer approval is at the end.

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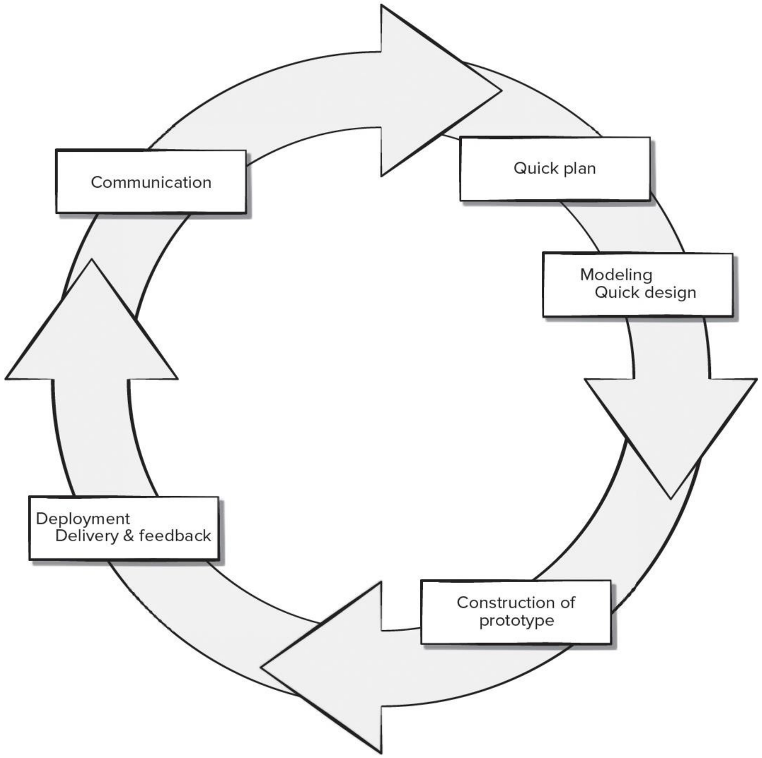
Prototyping Process Model



Pros

- Reduced impact of requirement changes.
- Customer is involved early and often.
- Works well for small projects.
- Reduced likelihood of product rejection.

Cons

- Customer involvement may cause delays.
- Temptation to "ship" a prototype.
- Work lost in a throwaway prototype.
- Hard to plan and manage.

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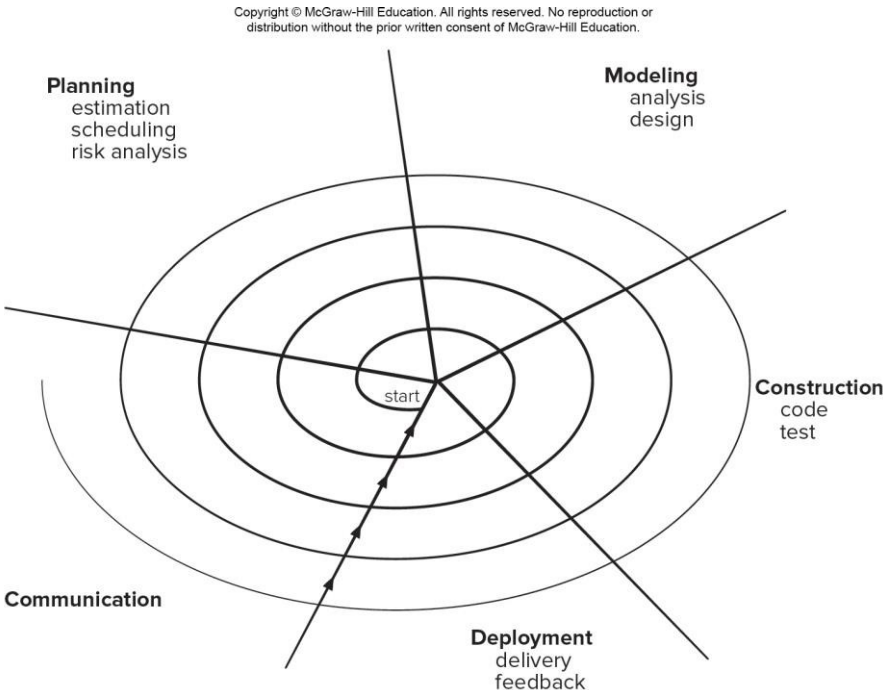
Spiral Process Model



Pros

- Continuous customer involvement.
- Development risks are managed.
- Suitable for large, complex projects.
- It works well for extensible products.

Cons

- Risk analysis failures can doom the project.
- Project may be hard to manage.
- Requires an expert development team.

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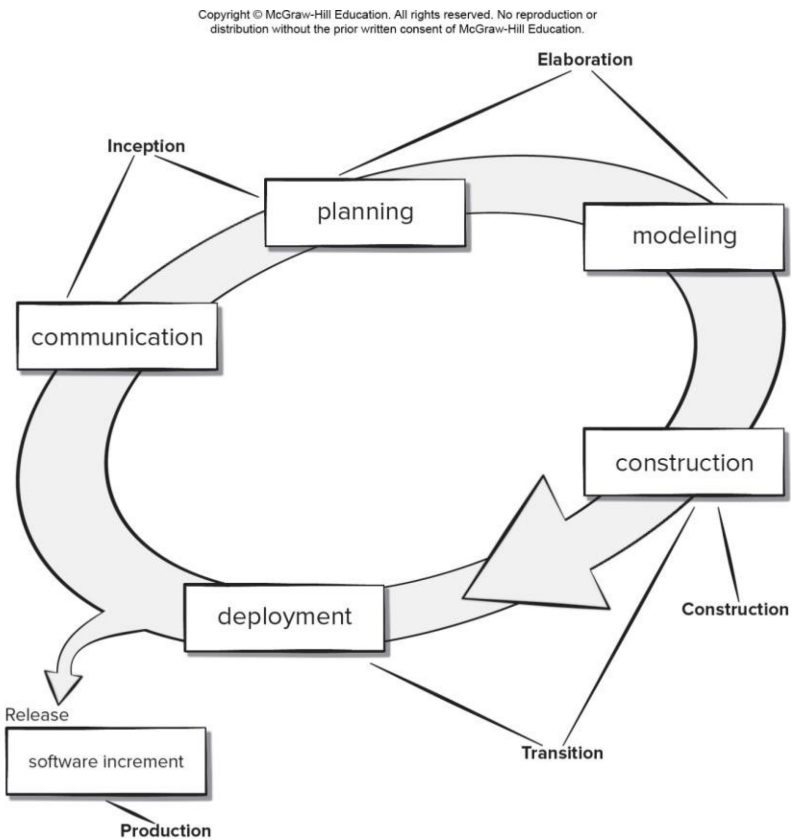
Unified Process Model



Pros

- Quality documentation emphasized.
- Continuous customer involvement.
- Accommodates requirements changes.
- Works well for maintenance projects.

Cons

- Use cases are not always precise.
- Tricky software increment integration.
- Overlapping phases can cause problems.
- Requires expert development team.

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Prescriptive Process Models 2

Prescriptive process models advocate an orderly approach to software engineering.

That leads to a two questions:

- If prescriptive process models strive for structure and order, are they appropriate for a software world that thrives on change?
- If we reject traditional process models and replace them with something less structured, do we make it impossible to achieve coordination and coherence in software work?

Question 1

A key difference between software and physical products, like hardware, is that software doesn't "wear out." What does this mean in practice?

A. Software never fails and has a zero failure rate over its entire lifespan.
B. Software failures are primarily not due to aging components but from undiscovered bugs or changes in its environment.
C. Software requires no maintenance once it is deployed to the customer.
D. The cost of maintaining and evolving software is always lower than the initial development cost.

Question 2

According to the prescriptive model characteristics, what is the primary purpose of defining a "task set" within a software process?

A. To guarantee that the final software product will be delivered on time and under budget.
B. To provide a rigid, unchangeable plan that the team must follow to ensure discipline.
C. To define the actual work to be done to accomplish the objectives of a software engineering action.
D. To replace the need for umbrella activities like quality assurance and risk management.

Homework

- Form teams of <=5 students each team and name your teams
- Select a real problem your team wants to resolve with Software
- Start discussing the problem within your team
- Provide the name of your team, team members and the selected problem to Teaching Assistants
- Send your questions from this class to Teaching Assistants